

Vickey Xia

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EDUCATION

BE (Hons) Computer Systems Engineering, University of Auckland; 2024 – 2027

- GPA: 8.06 / 9
- 2024 Dean's Honour list
- Modules: Electronics; Data Structures and Algorithms; Operating Systems; Control and Signal Processing Algorithms;

Personal Projects

Bluetooth DAC/AMP (Ongoing)

Modules: STM32, C, RTOS, BLE Audio, KiCad, Signal Processing, LTspice.

- Designing a battery-powered **Bluetooth LE 32-bit DAC**.
- MCU: **STM32WBA55CG** programmed in **C** with **CMSIS-RTOS**.
- **Custom BLE application** controlling Bluetooth connection, input DAC signals (**I2S**), OLED screen (**I2C**), rotary encoder, and buttons.
- Designed a **4-layer PCB** in **KiCad**, featuring QFN and TSSOP chips.
- Circuit design simulated and verified with **SPICE-Based** simulation.
- Status: Tested and verified prototype PCB using an **Oscilloscope**.

Ferrous Fellow (2 weeks)

Modules: RP2040, C++, C, AI/ML, Python.

- Created a toy robot using an **RP2040** programmed in **C** and **C++**.
- Speech and facial recognition: **TensorFlow Lite** and **OpenCV**.
- **OpenAI API** analyses emotions, which then recommends music.
- Implemented local TensorFlow, **I2C** for OLED screen, **UART** for device communication, **ADC** for microphone and **PWM** for servo motor control.

Custom Numeric Keypad

Modules: RP2040, C, KiCad, Autodesk.

- Designed a keypad with a **custom PCB** and 3D printed case.
- PCB was designed using **KiCad**, featuring an **RP2040** MCU.
- Firmware written in **C** utilising the QMK library.
- Case designed using **Autodesk Inventor** and **AutoCAD**.

LedRTOS

Modules: Real-Time Operating Systems, STM32, C, Assembly.

- Built from scratch an **RTOS** with C and Assembly on an STM32 chip.
- Implemented **scheduling** of LED blinks through **priority-based** pre-emptive scheduling with static priorities.

Gameboy / Chip8 Emulator

Modules: Hardware Emulation, C++, Unit Testing, SDL2.

- Emulated hardware of Game Boy and Chip-8 systems in **C++**.
- Catch2 used for **unit testing** framework to test CPU instructions.

SKILLS AND INTERESTS

• Coding Skills:

C, C++,
VHDL,
Python,
MATLAB,
Java,
R

• Software Tools:

KiCad, Altium,
LTspice, Cadence (PSpice),
Git, GitHub,
Autodesk (Inventor/AutoCAD),
Unit testing

• Hardware:

Soldering (SMT and THT),
Oscilloscope,
Digital Multimeter

• Languages:

English (native),
Mandarin (native),
Cantonese (native)

• Hobbies:

Hardware modification,
Audio signal processing,
Badminton,
Chess